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Published by IOP Publishing for Sissa Medialab

Received: *August 27, 2022*
Revised: *October 13, 2022*
Accepted: *October 18, 2022*
Published: *October 28, 2022*

Investigations on Complementary Split Ring Resonator (CSRR) array integrated printed conformal band pass filters for modern wireless communication applications

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Abstract: This paper presents the performance analysis of two different conformal band pass filters (BPFs) loaded with an array of circular complementary split ring resonators (CSRRs). Type-1 bandpass filter has been designed by integrating a 9×9 array of circular complementary split ring resonator on the bottom layer of the filter. Similarly, Type-2 bandpass filter has been constructed by loading an array of 5×5 split ring resonator on the patch layer of the filter. A comparative study between Type-1 and Type-2 bandpass filters are also evaluated that shows dual pass bands for Type-1 filter and triple pass bands for the Type-2 bandpass filter. Furthermore, the conformality of both bandpass filters are judged through bending deformation analysis with various bending positions at 15°, 30°, 45°, 60° and 90°. During the bending performance analysis, the suggested BPFs retains the dual and triple pass band characteristics without any major deviations in its characteristic's parameters. Various performance parameters like return loss, insertion loss, group delay, bandwidth, and quality factor are measured. Detailed analysis of E and H field distributions and surface current distributions are also presented. The Type-2 bandpass filter shows better performance in terms of filter properties compared to Type-1 band pass filter. Hence, the prototype of Type-2 filter with an area of $30 \times 30\,\text{mm}^2$ is fabricated and measured. The validated results are agreed with the simulated results which confirms the applicability of the proposed filter for several modern wireless communication applications such as Wi-Max and personal communication systems (PCS) applications.

Keywords: Attenuators, Filters; Passive components for microwaves

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https://doi.org/10.1088/1748-0221/17/10/P10043

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1 Introduction

The current trend of modern wireless communication systems necessitates the demands for compact system components and devices with multiband features to support multiple applications. In 20^{th} century, multiple wireless applications like mobile, wireless, and satellite communications are growing rapidly and each application are allowed to access limited frequency band spectrum. In this regard, high performance filters play a pivotal role in RF/Microwave circuitry for an effective utilization of allocated frequency spectrums for different wireless standards. For last few decades, filters implemented with microstrip technology have attracted the researchers of RF/Microwave community to a great extent. Even with the increasing demand of advanced miniaturized wireless devices, the precise design of microstrip bandpass filters with highly frequency selective nature are of great interest that can offer attractive performance with reduced cost and size. Bandpass filters with compact size, minimum cost and minimum insertion loss and high-performance impulse are highly required to support various wireless services [1, 2]. The main feature of bandpass filter is to allow only resonating frequency and suppress the undesired frequency bands [3]. Further, due to the demand of supporting multiple wireless applications, researchers have reported different dual band bandpass filters by using $\lambda/4$ resonator [4], stepped-impedance ring loaded resonator [5], T-shaped resonator [6], parallel coupled lines and shorted stubs [7] and open/shorted stubs [8]. Some triple-band bandpass filters are also designed and reported by utilizing various concepts such as tri-section SIRs [9], stepped-impedance ring resonator [10], stub loaded resonator [11, 12]. The reported traditional bandpass filters have been designed by integrating number of pairs of microstrip

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resonators which results in large size. The compact triple band bandpass filter design methods are adopted by optimizing the distance between two vias in the grounded stepped impedance resonator as reported in ref. [13]. Another compact triple band BPF was designed by J-M Yan et al. [14] by implementing multi-stub loaded resonator and asymmetrical SIRs. Another triple band miniaturized bandpass filter was realized with a floating U-shaped SIR and split ring resonator [15]. Furthermore, for the design methodology of BPF, it was demonstrated that electromagnetic characteristic of multi band can be strengthen by implementing multimode resonators [16, 17]. Individual control of bandwidth has been acknowledged by coupling impact [18] and modified quarter-mode substrate integrated waveguide (QMSIW) cavity [19].

Based on these discussed literature studies, the authors of this work got motivated to propose two different compact bandpass filter structures that can provide dual and triple pass bands by implementing metamaterials. Also, another initiated design objective is to make the designed filters conformal by maintaining dual and triple band characteristics. By the implementation of metamaterial structure due to its homogeneity condition, small size of the filter can be achieved [20]. Metamaterials are the structures with zero phase constant, negative refractive index, negative permittivity and negative permeability. The metamaterial structures are incorporated in the transmission line of the system with parallel inductance and capacitance and along with the series inductance and capacitance [21–24].

In this paper, two different compact and conformal bandpass filters have been designed using finite element method which includes an array of circular CSRRs in the designs. The conformal bandpass filters are designed using polyamide substrate. Comparative study between Type-1 and Type-2 BPFs are also evaluated in terms of all performance parameters. Apart from that, fabricated model of proposed Type-2 bandpass filter along with its measurement results are also presented.

The highlights of the proposed Type-2 filter are as follows:

i. The proposed bandpass filter occupies a compact dimension of $30 \times 30\,\mathrm{mm}^2$.

ii. The proposed filter geometry has been implemented with novel metamaterial-based array of circular complementary split ring resonators (CSRRs) structure.

iii. The proposed bandpass filter offers triple pass bands centered at 1.9 GHz, 3.8 GHz and 5.5 GHz.

iv. The proposed bandpass filter is suitable for multiple wireless communication applications such as Wi-MAX and PCS systems.

v. The flexible or bendable nature of the proposed conformal bandpass filter makes it suitable for the implementation in any conformal devices or can be placed over any type of mounting surfaces including curved structures.

vi. Performance of the proposed triple band BPF has been demonstrated with detailed analysis of return loss, insertion loss, group delay, bandwidth, quality factor, and field distributions.

2 Design specifications and procedures

2.1 The design of circular metamaterial structure

The circular CSRRs have been designed using etching of SRRs on the copper layer of the filter. The circular rings consist of radiating copper and have a minute gap in between them. A split ring resonator is embedded in the ground plane, corresponding to two-fold circular rings. On the metal surface the line structures with slots are demonstrated in figure 1. Moreover, CSRR is also assessed for accomplishing activity in the GHz band.

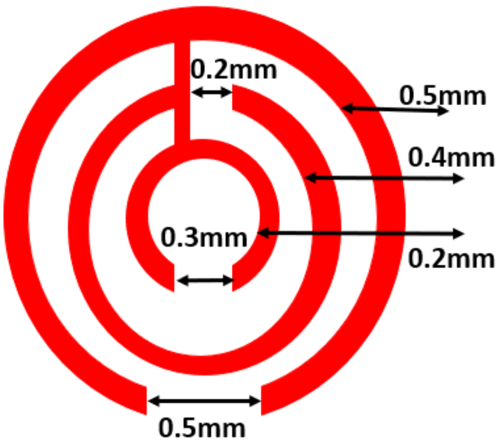

Figure 1. The dimension of the CSRR unit cell.

2.2 Metamaterial attributes

These metamaterials are defined as the structure which are man made by changing the structure of the existing elements and producing slits in the structure to produce negative mechanical characteristics. By the integration of these structure, the system exhibits high performance value it terms of its electromagnetic properties like return loss or insertion loss.

The metamaterial structure expresses the permittivity by utilizing following formula

$$\varepsilon_p = 1 - \frac{\omega_p^2}{\omega^2}, \tag{2.1}$$

Where 'ε_p' is the permittivity of the metamaterial structure 'ω_p' is plasma frequency, 'ω' frequency of the propagating electromagnetic wave.

Assuming the frequency below the plasma frequency, the productive permittivity is considered from the above condition. Figure 2 (a-b) states the permittivity value of the 9×9 array circular CSRR which resonates across 3.3 and 4.9 GHz. Similarly, figure 2 (c-e) states the permittivity value of the 5×5 array circular SRR which resonates across 1.9, 3.8 and 5.5 GHz.

The refractive index of the unit cell is analyzed by the following formulae

$$\begin{aligned} &\text{If } \mu < 0, \varepsilon < 0 \\ &\eta_{\text{eff}} = -\sqrt{\varepsilon\mu}, \end{aligned} \tag{2.2}$$

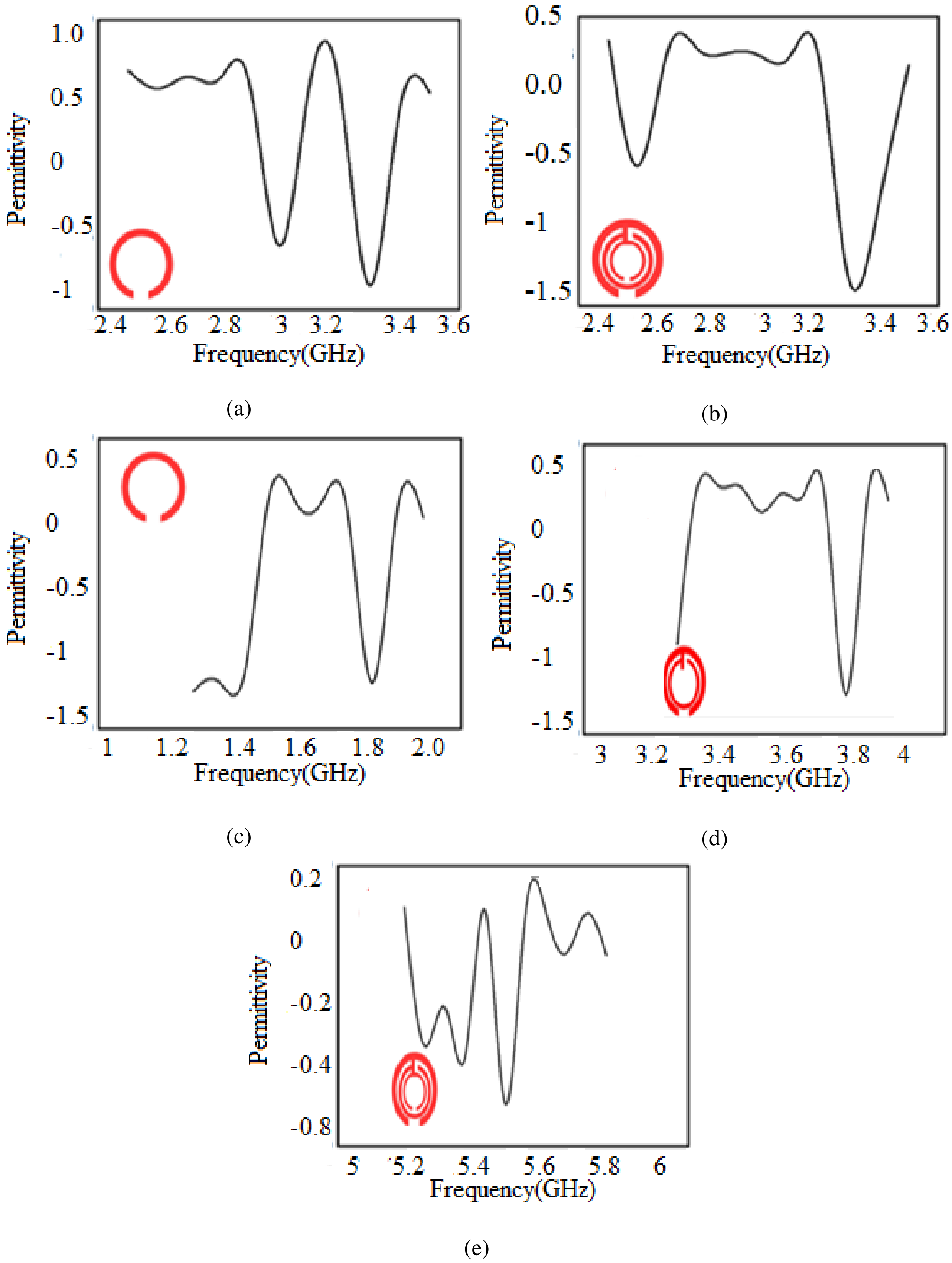

Figure 2. Frequency vs Permittivity (negativity permittivity) (a-b) 9×9 array circular CSRR structure and (c-e) 5×5 array circular SRR structure.

Where η_{eff} is the refractive index of the metamaterial, ε is the permittivity of the metamaterial and μ is the permeability of the metamaterial.

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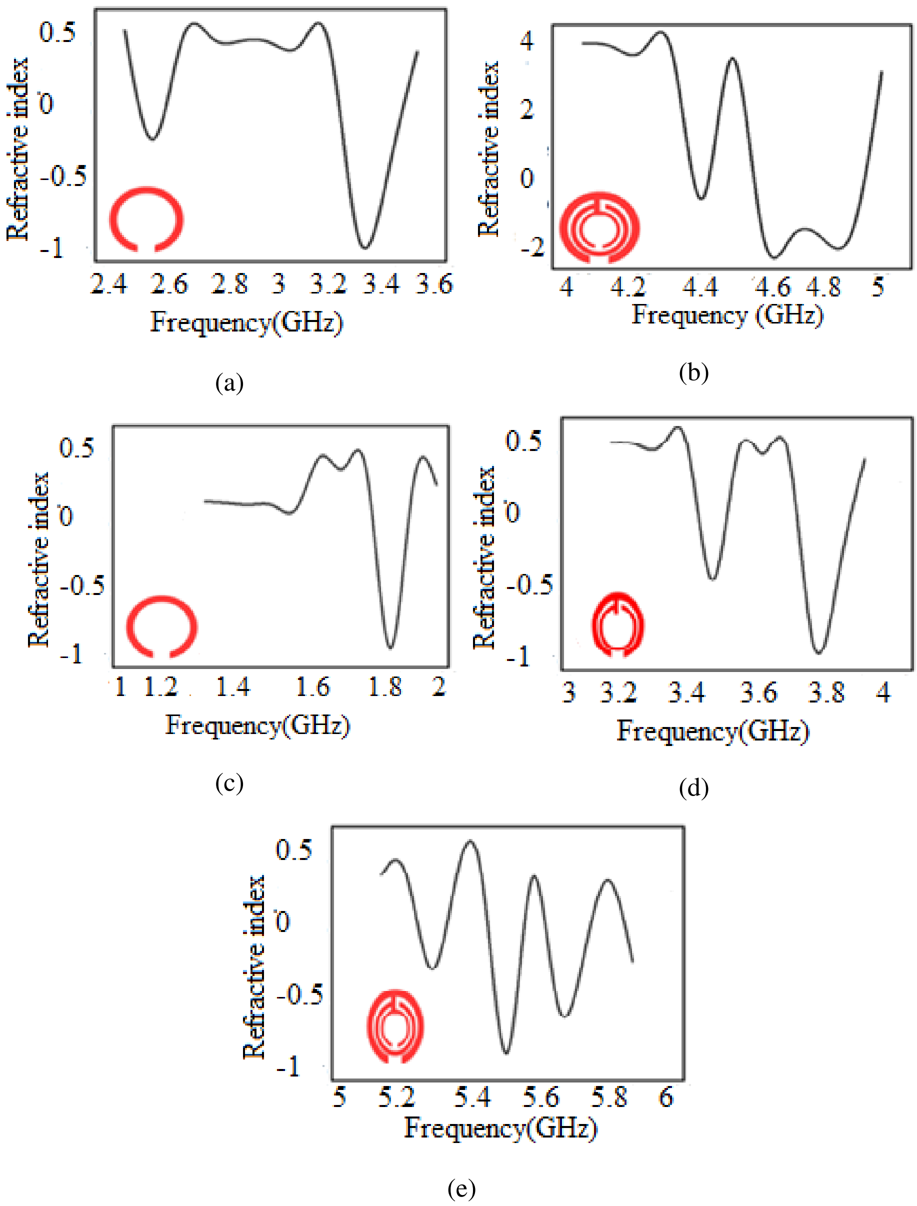

Figure 3. Frequency vs Refractive index curve (negative refractive index) (a-b) 9×9 array circular CSRRs (c-e) 5×5 array circular SRRs.

With the inclusion of additional rings figure 3 (a-b) states the refractive value of the 9×9 array circular CSRR which resonates across 3.3 and 4.9 GHz. Similarly, figure 3 (c-e) states the refractive index value of the 5×5 array circular SRR which resonates across 1.9, 3.8 and 5.5 GHz.

$$\mu_r = \frac{2\,(1 - V_1)}{\int G_0 d\,(1 - V_2)}, \tag{2.3}$$

Where μ_r is the permeability of the material, V_1 is $S_{11} = S_{21}$ and where V_2 is $S_{21} - S_{11}$, S_{11} is the return loss value, S_{21} is the insertion loss value, d is the thickness of substrate.

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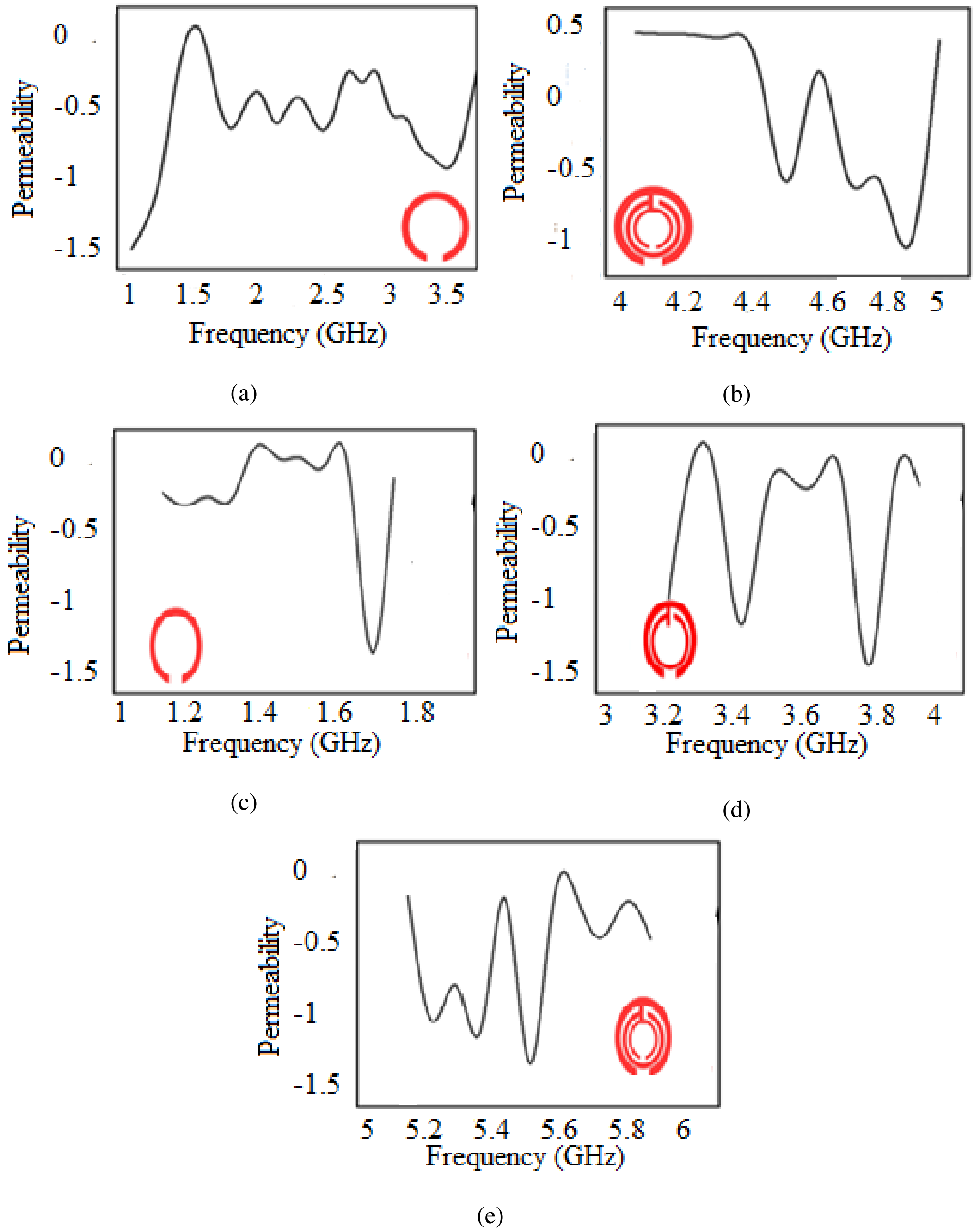


Figure 4. Frequency vs. permeability (a-b) (negative permeability) 9×9 array circular CSRR structure and (c-e) 5×5 array circular SRR structure.

The working frequency of Complementary Split ring resonator is by and large depend on the geometry of the CSRR. The frequency is relative with different times the outside diameter of CSRR. Figure 4 (a-b) states the Permeability value of the 9×9 array circular CSRR which resonates across 3.3 and 4.9 GHz. Similarly, figure 3 (c-e) states the Permeability value of the 5×5 array circular SRR which resonates across 1.9, 3.8 and 5.5 GHz.

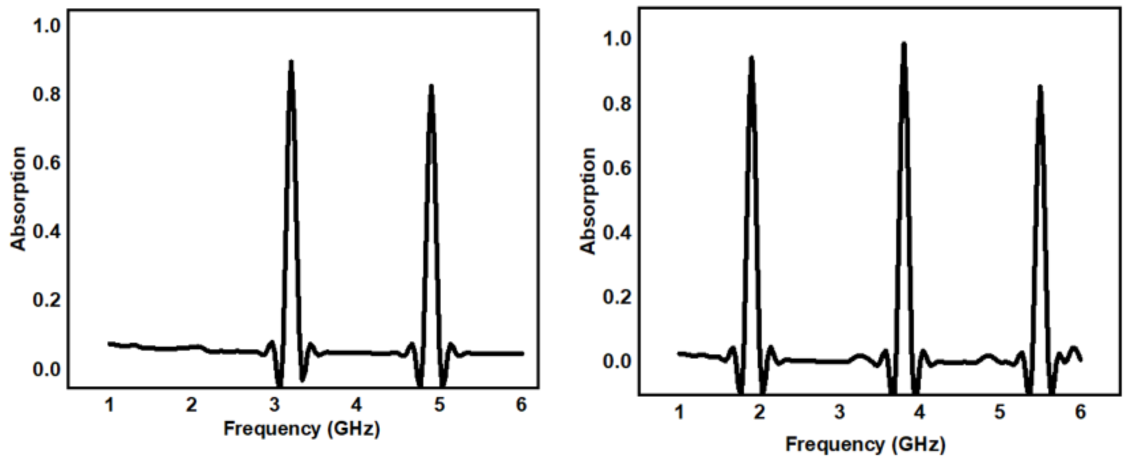
2.3 Absorption characteristics

For an advanced covertness execution, a unit cell safeguard with high ingestion and an enormous point relies upon electromagnetic resonance. Metamaterial absorber is associated the top or base layer of the filter to order the radar cross section.

Absorbability is evaluated by the given formula

$$\text{Absorption rate} = (1 - |S_{11}|^2 - |S_{21}|^2) \tag{2.4}$$

Where, $|S_{11}|$ is the return loss value of the bandpass filter and $|S_{21}|$ is the insertion loss of the bandpass filter. The 5×5 and 9×9 array of CSRR and SRR unit cell operates at 1.9, 3.8 and 5.5 GHz and 3.3 and 4.9 GHz various working frequencies by adding three circular rings. Here, high absorption value is obtained by the 5×5 SRR array compared to 9×9 CSRRs array.



(a) (b)

Figure 5. (a-b) Absorption value of 9×9 circular CSRR structure and 5×5 circular SRR structure.

2.4 Proposed bandpass filter with the integration of metamaterials

2.4.1 Type-1 bandpass filter

Figure 6 adresses the Type-1 BPF with a 9×9 array of Circular Complementary split ring resonator presented on the bottom layer of filter. The proposed BPF achieves impedance coordinating of 50 ohms. The metal rod over the top layer helps to accomplish proper impedance matching. The BPF has been designed on conformal polyamide substrate with a thickness of 0.1 mm. The proposed BPF achieves conformal properties at 15°, 30°, 45°, 60° and 90° providing wide coverage with miniaturization and inclusion of less space. The footprint of the BPF is $30 \times 30\ \text{mm}^2$. The dimension of the metal rod is of 30 mm length and width of 2 mm, respectively.

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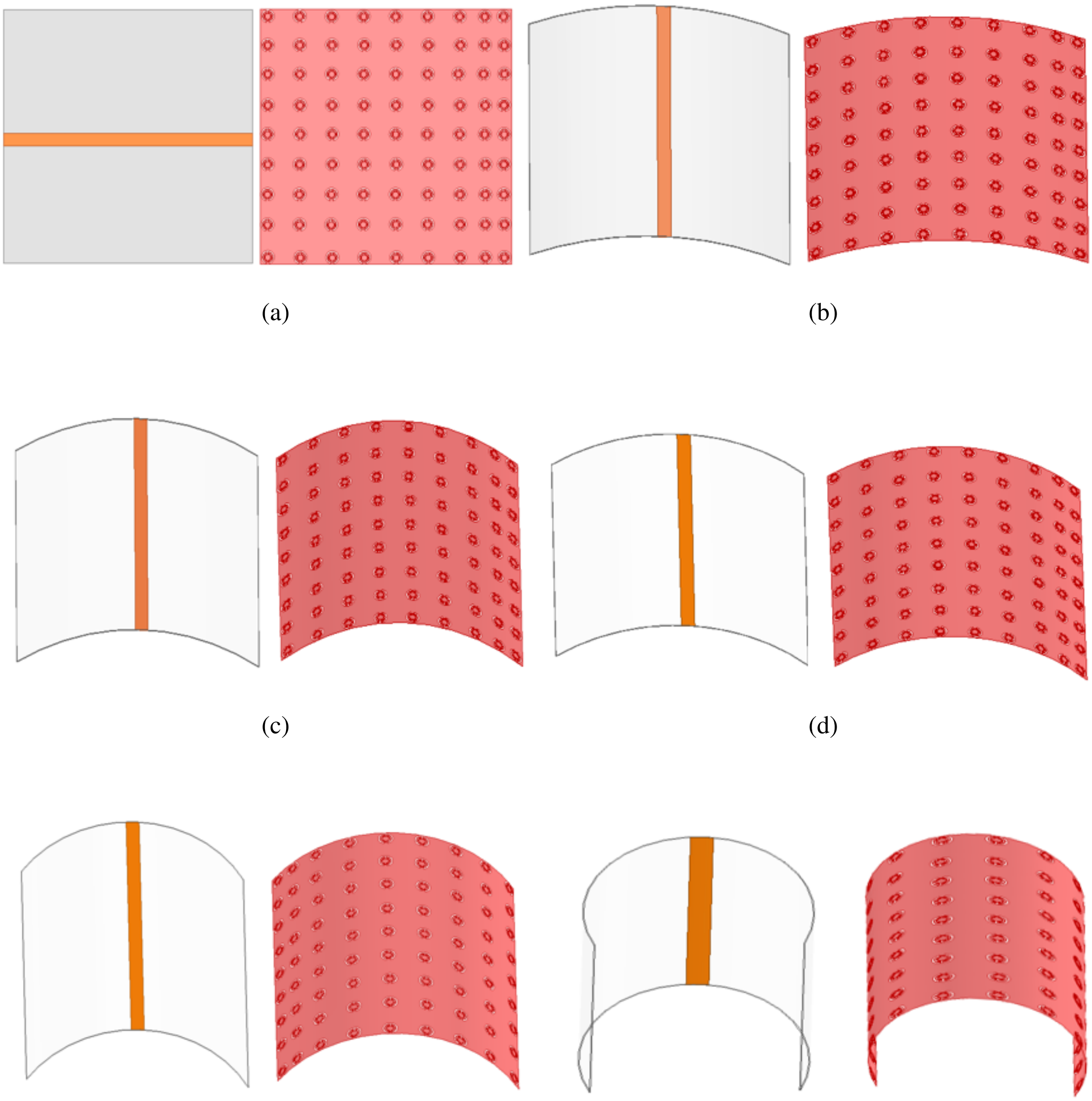

(a) (b) (c) (d) (e) (f)

Figure 6. Type-1 BPF (a) without bending and (b-f) with various bending angles at (15°, 30°, 45°, 60° and 90°).

2.4.2 Type-2 Bandpass filter

Figure 7 depicts the Type-2 BPF with full ground and an 5×5 array of SRR on the top most layer achieving an impedance coordination of 50 ohms. The design of the BPF is fabricated on a conformal substrate with a height of 0.1 mm. The proposed BPF achieve conformability at different angles of 15°, 30°, 45°, 60° and 90° results in wider bandwidth and miniaturization. The dimension of the BPF substrate with length of 30 mm and width of 30 mm. The geometry of the of the upper layer is 20 mm $\times$ 25 mm. The dimension of the feed line on the either side is of 2215 mm $\times$ 5 mm. The performance of both types of BPF are enhanced due to the implementation of SRRs.

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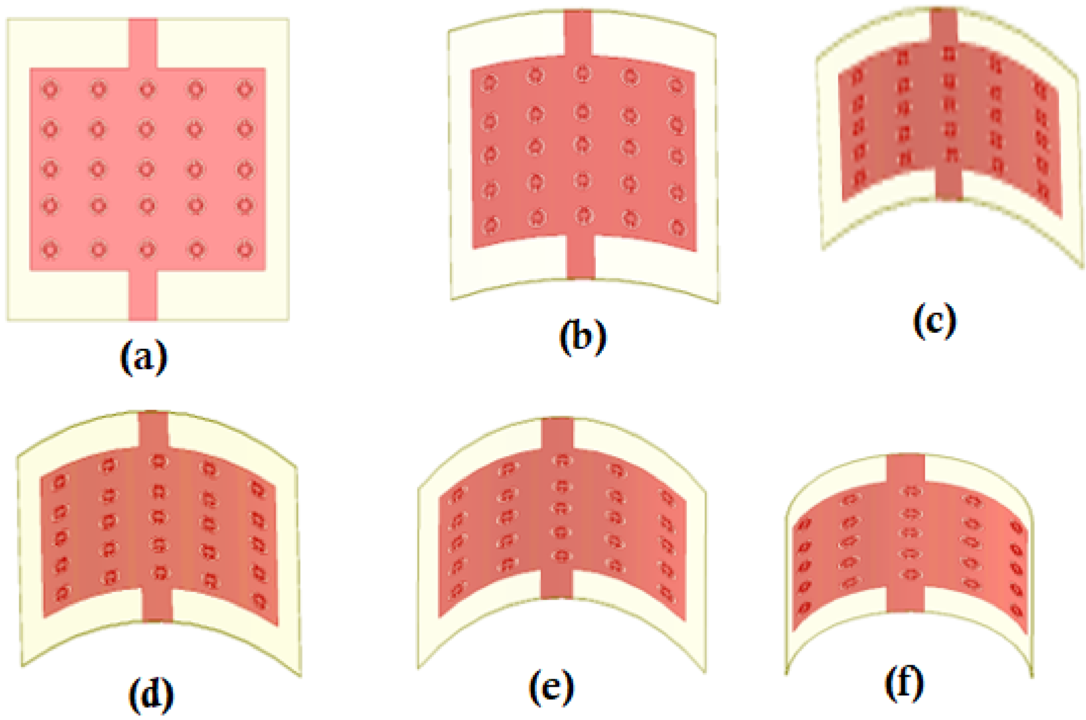

Figure 7. Type-2 BPF (a) without bending and (b-f) with various bending angles at (15°, 30°, 45°, 60° and 90°).

2.5 Passive element extraction

It is crucial to select the lumped parts for the bandpass filter. The lumped parts of BPF at different points is investigated. At first to survey the lumped parts, the order for the filter has been determined by Chebyshev technique. The filter has the ripple factor of 0.01 dB and with stop band distortion of 40 dB. The Type-1 BPF and Type-2 BPF tends to the order for 5. The equivalent circuit of both Type-1 and Type-2 bandpass filters is shown in figure 8.

The lumped components of the bandpass filter are attributed to the series inductance and capacitance values; and shunt inductance and capacitance values. These quantities are evaluated numerically using the equations [30, 31].

For series L_k is the inductance value and C_k is the capacitance value in eq. (2.5) and eq. (2.6)

$$L_k = \frac{L_{k^1} Z_0}{f_0 \Delta} \tag{2.5}$$

$$C_k = \frac{\Delta}{f_0 L_{k^1} Z_0} \tag{2.6}$$

For shunt L_k is the inductance value and C_k is the capacitance value in eq. (2.7) and eq. (2.8)

$$L_k = \frac{\Delta Z_0}{f_0 C_{k^1}} \tag{2.7}$$

$$C_k = \frac{C_{k^1}}{f_0 \Delta Z_0} \tag{2.8}$$

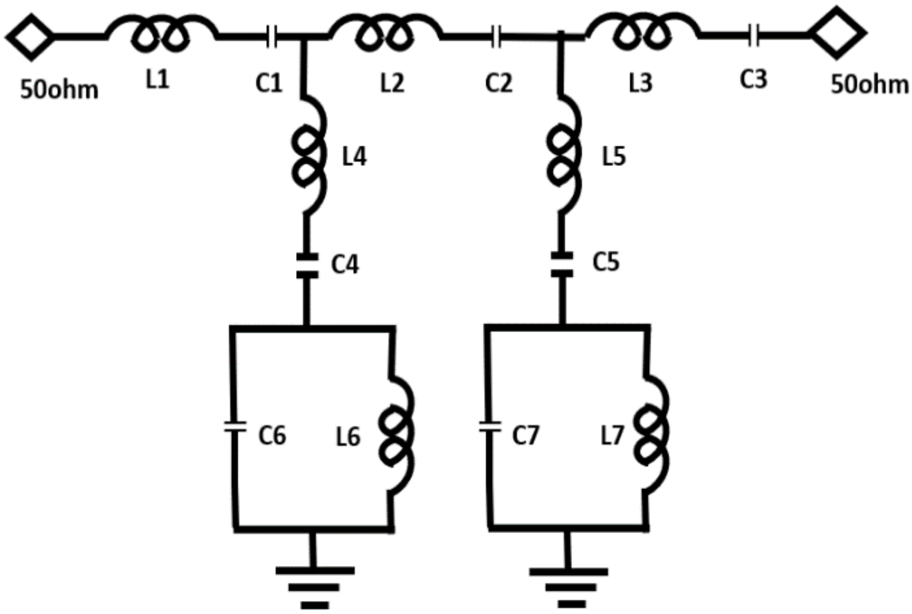

Figure 8. Equivalent circuit of both Type-1 and Type-2 bandpass filter. Where, L1 = L3, C1= C3, L4 = L5, C4 = C5, L6 = L7 and C6 = C7.

Where, Δ = fractional bandwidth and it can be evaluated by the following equation

$$\Delta = \frac{f_1 - f_2}{f_c} \tag{2.9}$$

Where f_1 is the higher cut off of frequency, f_2 is the lower cut off of frequency, f_c is the resonating frequency.

The L and C values for Type-1 BPF and Type-2 BPF are found for without bending and with bending at different bending points cases. Table 1 and 2 sumarrizes the lumped parts of the bandpass filters at different bending points (15°, 30°, 45°, 60° and 90°) along with non bending conditions.

Table 1. Lumped elements of Type-1 BPF.

Investigation	L1=L3 nH	C1=C3 pF	L2 nH	C2 pF	L4=L5 nH	C4=C5 pF	L6=L7 nH	C6=C7 pF
Without bending	27.95	0.04935	55.78	0.0384	56.78	0.4693	0.0469	17.31
With 15° bending	26.95	0.06935	55.78	0.0484	57.78	0.4693	0.0369	17.30
With 30° bending	22.95	0.06935	54.78	0.0684	56.78	0.7693	0.0869	19.31
With 45° bending	23.95	0.04935	55.78	0.0364	51.78	0.4693	0.0369	14.31
With 60° bending	24.95	0.07935	51.78	0.074	50.78	0.4693	0.0469	19.31
With 90° bending	25.95	0.04935	53.78	0.0484	51.78	0.7693	0.0969	18.31

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Table 2. Lumped elements of Type-2 BPF.

Investigation	L1=L3 nH	C1=C3 pF	L2 nH	C2 pF	L4=L5 nH	C4=C5 pF	L6=L7 nH	C6=C7 pF
Without bending	27.95	0.03935	59.78	0.0184	59.78	0.3693	0.0569	19.31
With 15° bending	26.95	0.02935	56.78	0.0284	54.78	0.2693	0.0269	16.31
With 30° bending	21.95	0.04935	56.78	0.0584	52.78	0.8693	0.0969	17.31
With 45° bending	23.95	0.05935	58.78	0.0164	54.78	0.2693	0.0169	15.31
With 60° bending	27.95	0.08935	54.78	0.084	51.78	0.5693	0.0969	18.31
With 90° bending	25.95	0.02935	51.78	0.0684	53.78	0.8693	0.0769	15.31

3 Fabricated prototype of Type-2 BPF and discussion on results

Figure 9 presents the fabricated prototype of the proposed Type-2 bandpass filter.The phtographs of the fabricated model of the Type-2 BPF are displayed for without bending and conditions. The variations in return losses ($|S_{11}|$) for the case of without bending and with bending analysis at different bending positions are experimentally analyzed as depicted in figures 10 (a) and (b), for Type-1 and Type-2 BPF filters, respectively. From the results, it is observed that as the curvature of the surface changes, the resonant frequency also changes. The insertion losses for the same conditions are shown in figure 11 (a) and (b) for Type-1 and Type-2 bandpass filters. By integrating the Split ring resonator and Complementary Split ring resonator unit cells to the band pass filter, the characteristic parameters of the filter are enhanced. It can be observed from figure 10 (a) that Type-1 bandpass filter shows the return loss value variations in between −30 dB to −18 dB for the cases of without bending and with bending (15, 30, 45, 60, and 90 degrees) across the operating frequencies. From figure 10 (b), it is observed that Type-2 bandpass filter shows the return loss value in between −34 dB to −20 dB for without bending and with bending at 15, 30, 45, 60, and 90 degrees across the operating frequencies centered at 1.9, 3.8, and 5.5 GHz. From figure 10, it can be seen that Type-2 bandpass filter presents great execution than Type-1. Type-2 performs well at one of the three relating triple band working frequencies. The above frequency includes Wi-Max and Personal communication systems (PCS) applications.

As depicted in figure 11 (a), the insertion losses for the Type-1 bandpass filter varies in between −2.3 dB to −1.1 dB considering the cases for without and with bending from 0 to 90 degree across the operating frequencies. Similarly, it can be seen from figure 11 (b) that Type-2 bandpass filter shows the insertion loss value in between −2.45 dB to −0.6 dB for without and with bending from 0 to 90 degree across the operating frequencies.

From the findings, it can be concluded that Type-2 bandpass filter offers better performance than Type-1 filter. The proposed filter can be utilized for different wireless communication applications namely PCS and Wi-MAX. Consequently, the proposed flexible bandpass filter can be easily mounted on any curved surfaces due to its conformal feature as verified from bending analysis. Due to the conformal nature, the designed BPF assures further area miniaturization without degradation in its performance. So, the proposed BPF (Type-2) offers improved performance due to the execution of SRRs in its design.

Figure 12 indicates that the proposed Type-1 BPF attains minimum group delay in the range of 0.1–1.4 ns whereas Type-2 BPF achieves 0.1–1.9 ps group delay. From the observation, it is clear

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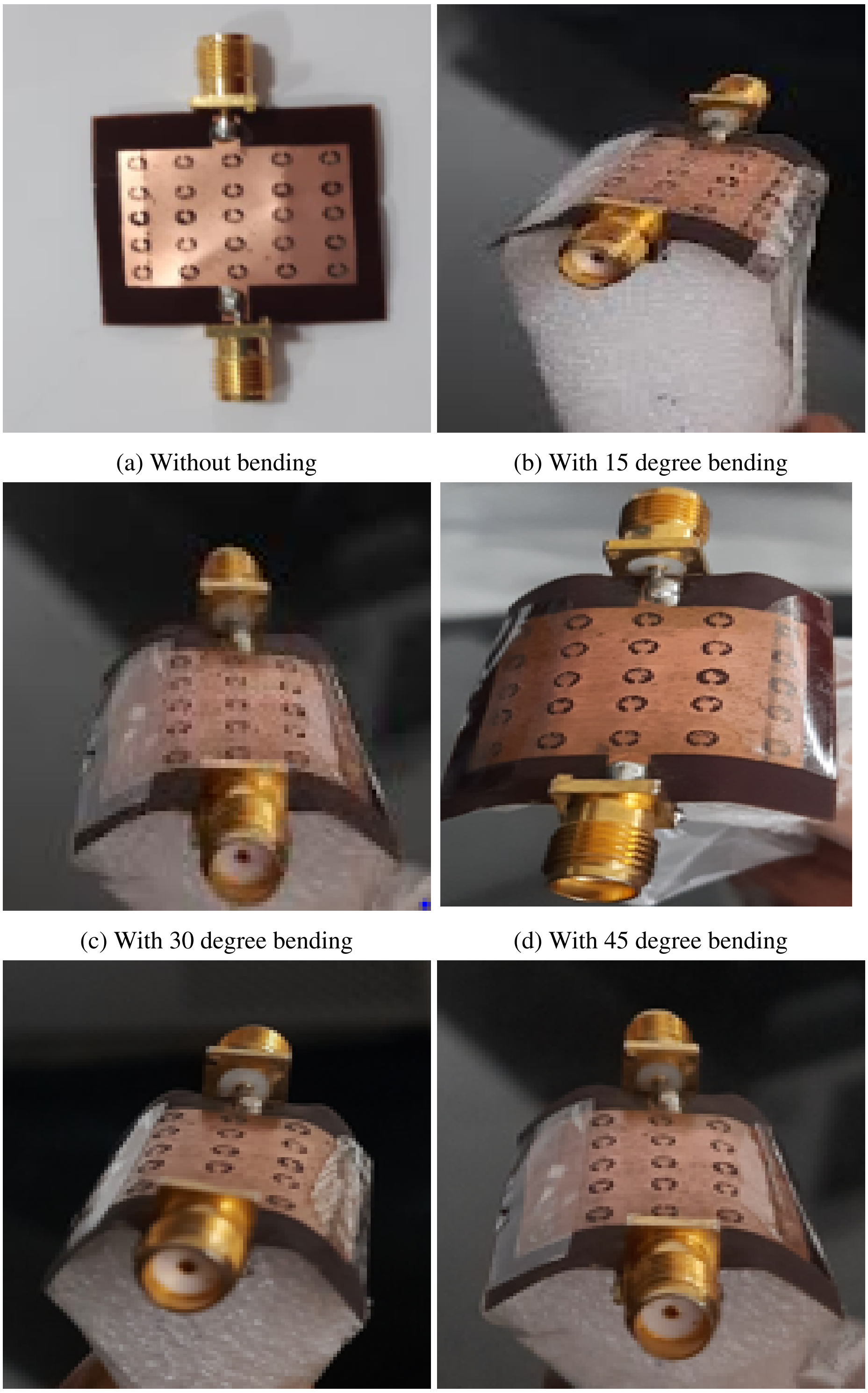

(a) Without bending (b) With 15 degree bending

(c) With 30 degree bending (d) With 45 degree bending

(e) With 60 degree bending (f) With 90 degree bending

Figure 9. Prototype model of proposed Type-2 bandpass filter (a-b) without bending and with 15 degree bending, (c-d) with 30 degree bending and with 45 degree bending, (e-f) with 60 degree bending and with 90 degree bending.

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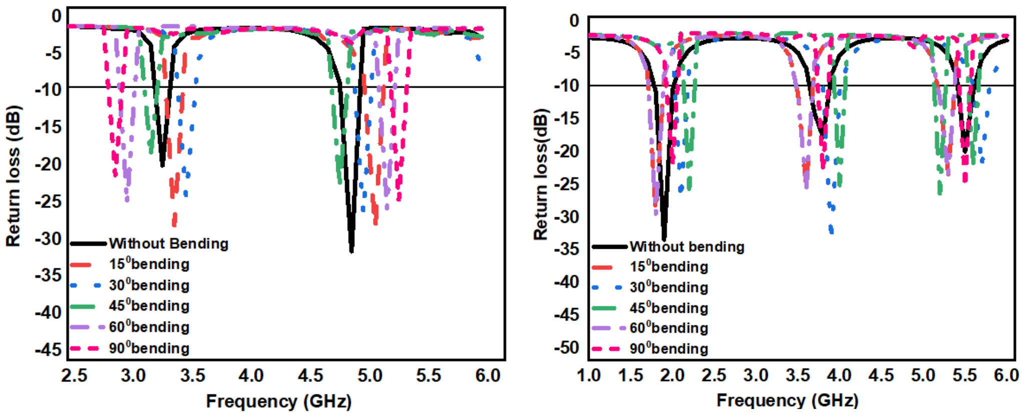
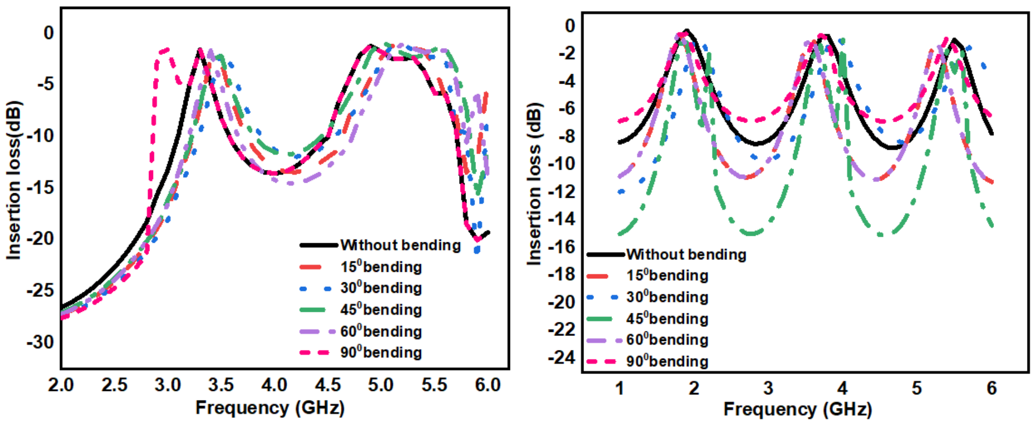


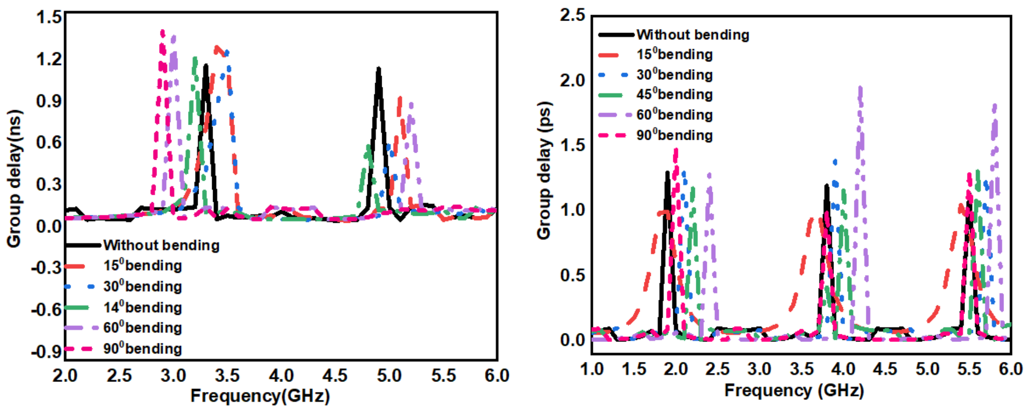
(a) (b)

Figure 10. Return loss values of designed bandapass filters (a) Type-1 BPF (b) Type-2 BPF.



(a) (b)

Figure 11. Insertion loss values of designed bandpass filters (a) Type-1 BPF (b) Type-2 BPF.



(a) (b)

Figure 12. Group delay of (a) Type-1 BPF and (b) Type-2 BPF for without and with bending analysis.

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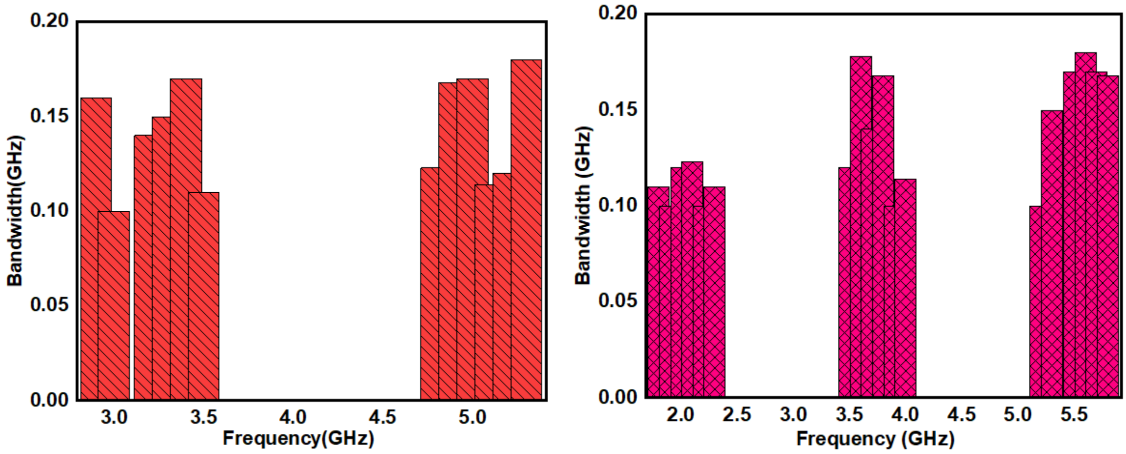
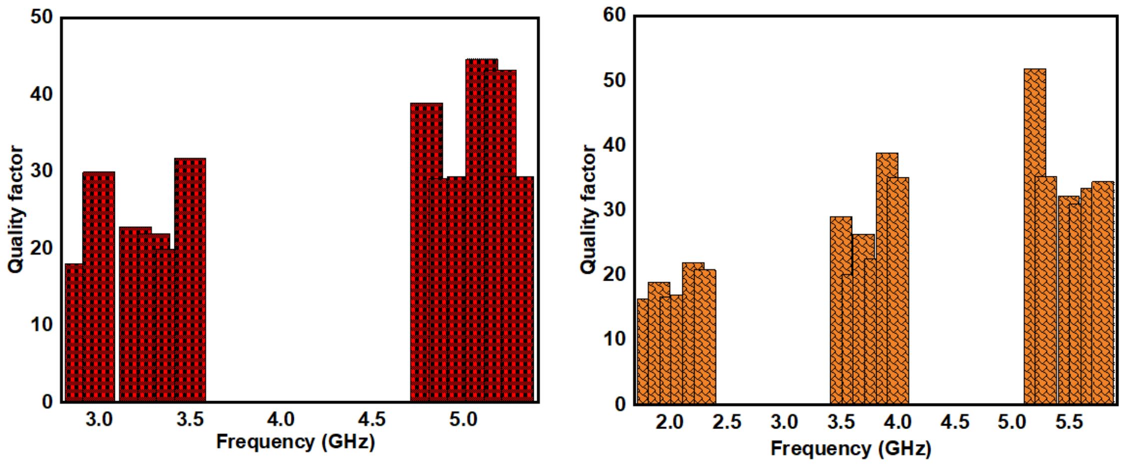


(a) (b)

Figure 13. Measured Bandwidth of (a) Type-1 BPF and (b) Type-2 BPF.



(a) (b)

Figure 14. Quality factor analysis of (a) Type-1 BPF and (b) Type-2 BPF.

that Type-2 BPF offers a better performance than Type-1 BPF in terms of group delay performance. Figure 13 presents the measured bandwidth of the proposed bandpass filters. The Type-1 BPF is realized with less bandwidth. Type-2 BPF attains good performance with improved transmission capacity which finally results in enhanced quality factor.

The quality factor analysis for Type-1 and Type-2 bandpass filters are presented in figure 14. It is clear that Type-2 BPF performs good execution compared to Type-1 filter. The assessment of quality factor is important for any device to evaluate the performance of the device. Obviously higher the quality factor, the performance of the filter will likewise be better.

As the bending point approaches, the boundaries like electric field, magnetic field, and surface current flows are marginally declining. The reduction of the mentioned parameters are due to alterations in surface angle as observed in figure 15. Red colour indicates the maximum field distribution i.e (electric, magnetic and surface currents) and blue colour indicates the minimum field

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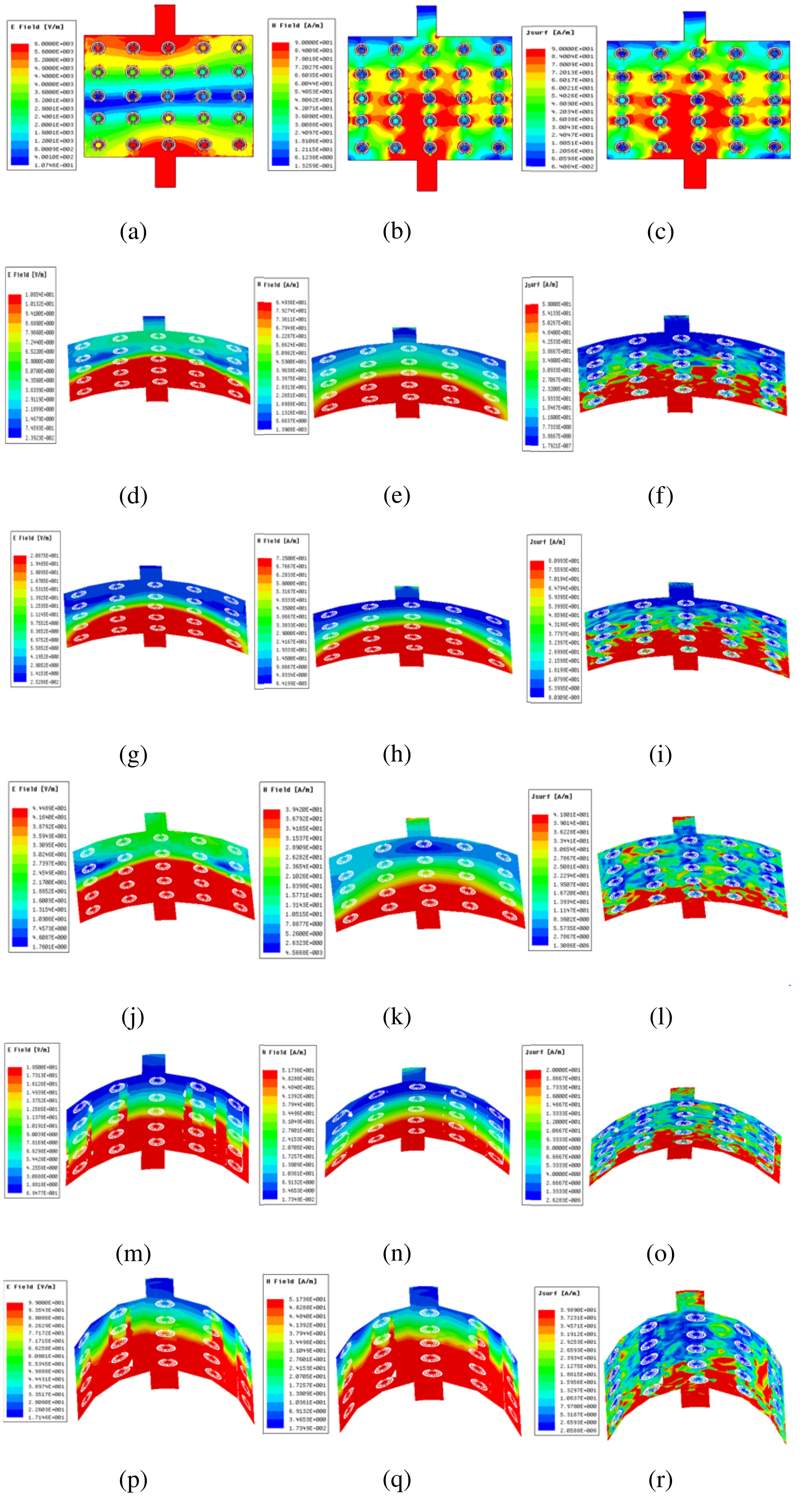

Figure 15. Electric, magnetic fields and surface current distribution of proposed Type-2 BPF at (a-c): non bending and (d-f): 15°, (g-i): 30°, (j-l): 45°, (m-o): 60°, and (p-r): 90° bending angles.

Table 3. Parameters of Type-1 BPF.

Bending angles	Filter order	Working frequency (GHz)	S_{11} (dB)	S_{12} (dB)	Group delay (ns)	B.W (GHz)
Without bending	5	3.3 4.9	−18.7 −30.3	−1.5 −1.1	1.11 1.0	0.15 0.168
With 15° bending	5	3.4 5.1	−26.8 −27.3	−2.4 −1.19	1.24 0.88	0.17 0.11
With 30° bending	5	3.5 5	−22.9 −25	−1.6 −2.6	1.23 0.58	0.11 0.17
With 45° bending	5	3.2 4.8	−17.3 −21.6	−1.8 −1.6	1.16 0.55	0.14 0.123
With 60° bending	5	3 5.2	−23.8 −24.5	−1.4 −1.0	1.36 0.84	0.1 0.12
With 90° bending	5	2.9 5.3	−20 −23.3	−1.9 −2.3	1.38 0.9	0.16 0.18

distribution. The maximum field distirbutions i.e (electric, magnetic and surface currents) ensures minimum losses in the designed filter. Table 3 states different parameter values of the Type-1 BPF across various bending angles and without bending angle. The Type-1 BPF operates across Wi-MAX and WLAN applications. Similarly, table 4 presents the characteristic parameters of the proposed Type-2 BPF including return loss, insertion loss, group delay and bandwidth for various bending angles and also without bending angle. The proposed Type 2 BPF operates at 1.9 GHz personal communication systems (PCS) and 3.8/5.5 GHz Wi-MAX wireless communication applications.

4 Performance comparison analysis with other proposed BPF filters

The proposed bandpass filter exhibits good performance when compared with some previous literature works [25–29] as shown in table 5. The filter parameters are analyzed for both filters. The proposed filter (Type-2 BPF) achieves good quality factor compared with the previous works. The proposed filter achieves miniaturization and conformability considering with the polyimide material.The fabricated bandpass filter showing compact size, with high performance compare with the exsisting literatre survey. By the implementation of the flexible substrate bending analysis will not add any additional band except slight disturbance in resonant band and sometime change in operating band as per the orientation of current distribution. The optimized design is suitable at any critical condition even at, bending, rolling will not affect the performance characteristics with much more variations.

5 Conclusion

With the implementation of the split ring resonator in the microstrip bandpass filter, the performance efficiency of the filter is increased, and they are applicable for various application like mobile, WiMAX and WLAN. Both the designed Type-1 and Type-2 BPFs shows the return loss value below

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Table 4. Type-2 BPF parameters.

Banding angles	Filter order	Working frequency (GHz)	S_{11} (dB)	S_{12} (dB)	Group delay (ns)	B.W (GHz)
Without bending	5	1.9 3.8 5.5	−31.5 −15.8 −18	−0.8 −0.6 −0.9	1.29 1.19 1.13	0.1 0.16 0.17
With 15° bending	5	1.8 3.6 5.3	−27.6 −38 −21.5	−1.5 −0.9 −1.2	0.98 0.9 0.74	0.11 0.178 0.15
With 30° bending	5	2.1 3.9 5.1	−25.5 −23.7 −22.5	−0.7 −1.3 −1.7	1.34 1.38 1.28	0.123 0.15 0.114
With 45° bending	5	2.2 4 5.2	−23.8 −23.2 −24.5	−1.9 −0.9 −1.2	1.17 1.15 1.3	0.1 0.14 0.1
With 60° bending	5	1.7 3.7 5.6	−27.6 −24.4 −20.0	−1.5 −2.7 −1.5	1.28 1.94 1.84	0.11 0.14 0.18
With 90° bending	5	2 3.5 5.8	−20.7 −20.4 −20.34	−1.9 −2.4 −1.4	1.49 1.0 1.3	0.12 0.14 0.168

Table 5. Comparison of the proposed BPF with other published works. (NA = Not Analyzed.)

Reference	Operating frequency (GHz)	Return loss (dB)	Insertion loss (dB)	Size (mm^3)	Bandwidth (GHz)	Group delay (ps)	Conformality	Applications
[25]	6.8	< 30	< 3	$30 \times 30 \times 0.81$	3.9–4.3	NA	NA	Wireless
[26]	9.95	14.14	0.75	$20.33 \times 20.33 \times 0.81$	NA	NA	NA	Satellite
[27]	5 to 9.5	< 40	< 3	$30 \times 30 \times 0.5$	NA	NA	NA	Wireless
[28]	2.4, 5.2	< 15	< 3	$1 \times 1.2 \times 1.6$	NA	NA	NA	WLAN
[29]	10.2, 16.2	< 15	< 3	$7 \times 7 \times 1.6$	6 and 6.6	NA	NA	Ku-band
Proposed work	1.9, 3.8, 5.5	−31.5, −15.8, −18	−0.8, −0.6, −0.9	$30 \times 30 \times 0.1$	0.1, 0.16, 0.17	1.29, 1.19, 1.13	YES	Mobile/ Wi-Max/ WLAN

−15 dB across the operating frequencies. By the implementation of the flexible suubstrate the bending will change the orientation of current distribution over the surface of the filter and which will affect the change in reflection co-efficient and other parameters in our models. The bending analysis for the proposed designs proved that, the performance characteristics not altered much and the same resonant band with slight variation in the level only observed. A comparative study between Type-1 and Type-2 bandpass filter is presented and it can be concluded that Type-2 bandpass filter

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demonstrates overall much better execution than Type-1 bandpass filter. By the implementation of the conformability, the proposed Type-2 bandpass filter resonates at various frequency bands which can be used for modern wireless communication applications (Wi-Max and PCS).

Acknowledgments

The authors would like to thank DST for the technical support through SR/FST/ET-II/2019/450.

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